Assignment 1

Data Mining and Analytics (SRM Institute of Science and Technology)

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Apply: (i) Apriori Algorithm with Candidate Generation
(ii) without candidate Generation [FP Growth]
(iii) with vertical data format

for following transactions:

TID	ITEMS
1	I_1, I_2, I_5
2	I_2, I_4
3	I_2, I_3
4	I_1, I_2, I_4
5	I_1, I_3
6	I_2, I_3
7	I_1, I_3
8	I_1, I_2, I_3, I_5
9	I_1, I_2, I_3

Min. support = 22%

Min. Confidence = 70%

$$\text{Min-Support} = \frac{22}{100} \times 9 \Rightarrow \frac{198}{100} \Rightarrow 1.98$$

$$\Rightarrow \simeq 2$$

So, $\boxed{\text{Min-Support} = 2}$

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(i) APRIORI ALG:

1-ITEMSET

ITEM	SUP_COUNT	
I_1	6	≥ 2
I_2	7	≥ 2
I_3	6	≥ 2
I_4	2	≥ 2
I_5	2	≥ 2 ✓

2-ITEMS TET

ITEM	SUP_COUNT	
I_1, I_2	4	≥ 2 ✓
I_1, I_3	4	≥ 2 ✓
I_1, I_4	1	< 2 ×
I_1, I_5	2	≥ 2 ✓
I_2, I_3	4	≥ 2 ✓
I_2, I_4	2	≥ 2 ✓
I_2, I_5	2	≥ 2 ✓
~~I_3, I_4~~	0	< 2 ×
~~I_3, I_5~~	1	< 2 ×
~~I_4, I_5~~	0	< 2 ×

3-ITEMSET

ITEM	SUP_COUNT	
I_1, I_2, I_3	2	≥ 2 ✓
I_1, I_2, I_4	1	< 2
I_1, I_2, I_5	2	≥ 2 ✓

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Association Rule:

For: I_1, I_2, I_3 (Confidence Value)

$\{I_1, I_2\} \rightarrow \{I_3\} \Rightarrow 2/4 \times 100 \Rightarrow 200/4 \Rightarrow 50\%$

$\{I_1, I_3\} \rightarrow \{I_2\} \Rightarrow 2/4 \times 100 \Rightarrow 200/4 \Rightarrow 50\%$

$\{I_2, I_3\} \rightarrow \{I_1\} \Rightarrow 2/4 \times 100 \Rightarrow 200/4 \Rightarrow 50\%$

All the values are less than Min. Confidence (ie) 70%.

Hence, it is not a strong Rule.

For I_1, I_2, I_5 (Confidence Value)

$\{I_1, I_2\} \rightarrow \{I_5\} \Rightarrow 2/4 \times 100 = 50\%$ ✗

$\{I_1, I_5\} \rightarrow \{I_2\} \Rightarrow 2/2 \times 100 = 100\%$ ✓

$\{I_2, I_5\} \rightarrow \{I_1\} \Rightarrow 2/2 \times 100 = 100\%$ ✓

The Association

$\{I_1, I_5\} \rightarrow \{I_2\}$ and $\{I_2, I_5\} \rightarrow \{I_1\}$

are strong rule.

(ii) FP GROWTH

1-ITEMSET

ITEM	SUP_COUNT	
I_1	6	≥ 2 ✓
I_2	7	≥ 2 ✓
I_3	6	≥ 2 ✓
I_4	2	≥ 2 ✓
I_5	2	≥ 2 ✓

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List L = $\{I_2 : 7, I_1 : 6, I_3 : 6, I_4 : 2, I_5 : 2\}$

TID	ITEMS
1	$\{I_2, I_1, I_5\}$
2	$\{I_2, I_4\}$
3	$\{I_2, I_3\}$
4	$\{I_2, I_1, I_4\}$
5	$\{I_1, I_3\}$
6	$\{I_2, I_3\}$
7	$\{I_1, I_3\}$
8	$\{I_2, I_1, I_3, I_5\}$
9	$\{I_2, I_1, I_3\}$

FP - TREE ROOT NODE $\{\} \rightarrow$ NULL

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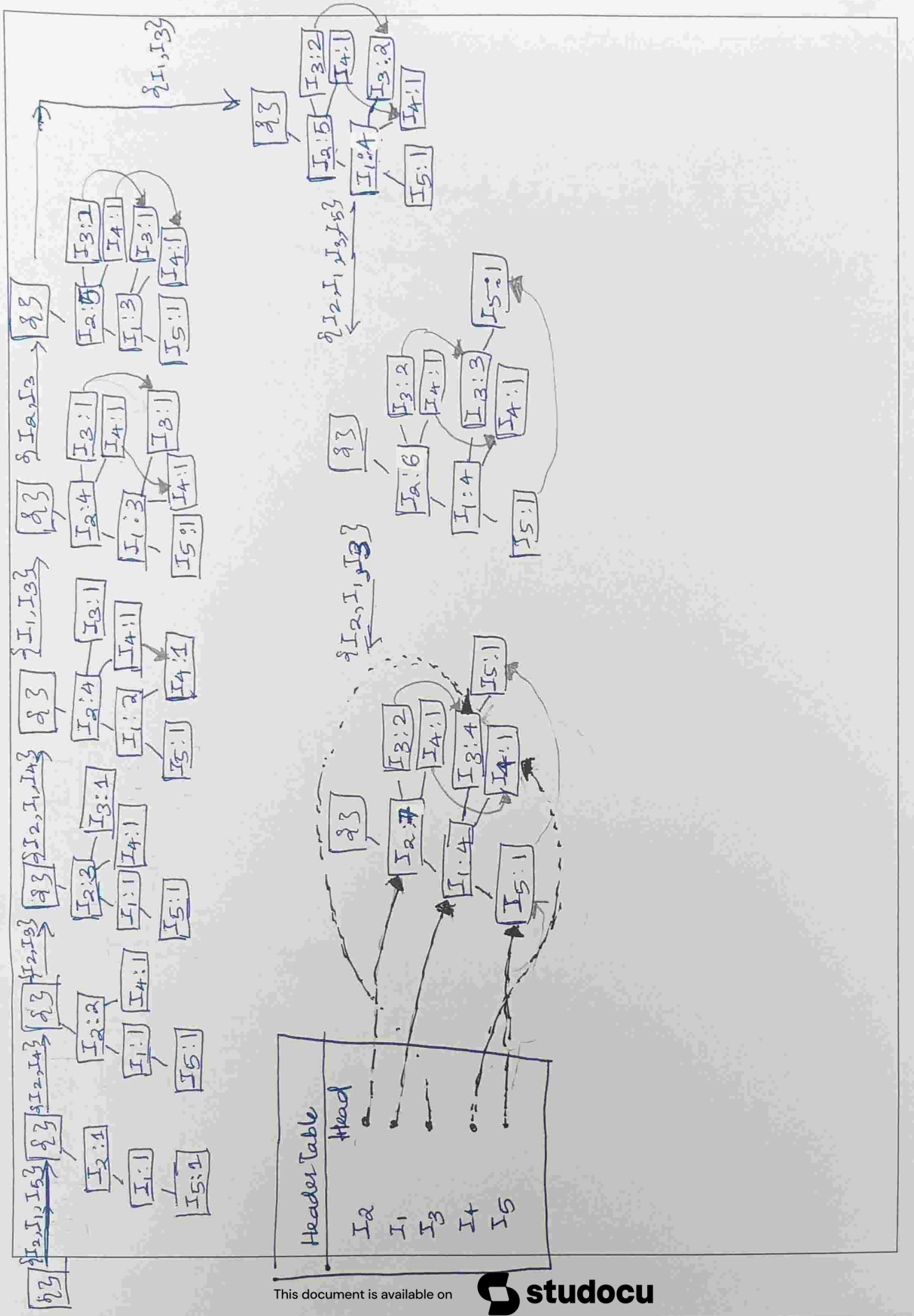

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(iii) Vertical Data Format

1-ITEMSET

ITEMS	TID	
I_1	{1,4,5,7,8,9}	= 6 ✓
I_2	{1, 2, 3, 4, 6, 8, 9}	= 7 ✓
I_3	{3, 5, 6, 7, 8, 9}	= 6 ✓
I_4	{2, 4}	= 2 ✓
I_5	{1, 8}	= 2 ✓

≥ 2

2-ITEMSET

ITEMS	TID
I_1, I_2	{1, 4, 8, 9} = 4 ✓
I_1, I_3	{5, 7, 8, 9} = 4 ✓
I_1, I_4	{4} = 1 ×
I_1, I_5	{1, 8} = 2 ✓
I_2, I_3	{3, 6, 8, 9} = 4 ✓
I_2, I_4	{2, 4} = 2 ✓
I_2, I_5	{1, 8} = 2 ✓
I_3, I_4	{ } = 0 ×
I_3, I_5	{8} = 1 ×
I_4, I_5	{ } ⇒ 0 ×

3-ITEMSET

ITEMS	TID	
I_1, I_2, I_3	{8, 9}	= 2 ✓
I_1, I_2, I_4	{4}	= 1 ×
I_1, I_2, I_5	{1, 8}	= 2 ✓

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- Hence, the candidate generation process generates only two, 3-Itemsets:

$$\{I_1, I_2, I_3\} \text{ and } \{I_1, I_2, I_5\}.$$

- Also, the ~~two~~ "3-itemsets" $\{I_1, I_2, I_3\}$ and $\{I_1, I_2, I_5\}$ are frequent Itemsets.

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